

LOGBOOK: EXTREMELY LONG-FORM AUDIO EVENT UNDERSTANDING

Kwanghee Choi^{*,*}, Suwon Shon^{*}, Dmitriy Serdyuk^{*}, Guitang Lan^{*}, Chao-Wei Huang^{*},
 Mohammad Sadegh Rasooli^{*}, Sangeeta Srivastava^{*}, Zhaojiang Lin^{*}, Saurabh Adya^{*}, Ming Sun^{*}

^{*}UT Austin, USA ^{*}Meta Reality Labs, USA

ABSTRACT

Audio benchmarks are built around short, pre-segmented clips, limiting model design to brief inputs or fixed vocabularies. To close this gap, we introduce LOGBOOK, a benchmark for hour-scale audio understanding, with recordings ranging from ten minutes to six days. Given a continuous audio recording and an event label vocabulary, a system must predict a gap-free segmentation with an event label and a description per segment. We compare 52 systems, end-to-end and cascaded, and ablate fine-tuning, context length, and reasoning budget. We find the task tractable, though the best systems remain below the human reference. Also, over-segmentation is pervasive, and fine-tuning partially mitigates it. Finally, end-to-end are often better than cascaded systems, but degrades with longer context.¹

Index Terms— long audio, audio event segmentation, egocentric audio, audio-language model, benchmark.

1. INTRODUCTION

Hearing is continuous. Throughout the day, we experience an unbroken acoustic stream and parse into segments such as conversation, meal, or commute, where wearable devices are now able to capture [1, 2]. However, existing audio event datasets often consist of short clips, already cut to contain a single event [3–5]. Some egocentric datasets provide continuous records [6, 7], but none is designed for long audio: many often run under an hour [8–11], cover only a specific activity such as kitchen work [12, 13], or treat audio as secondary to video [6, 7] (see [7] for a survey). Finally, long-audio benchmarks reach for duration, yet skew toward speech [14], and capped at 20 and 30 minutes [15, 16].

The gap influences model design. Sound event detectors are often built and evaluated on short recordings with a fixed vocabulary [17]. Audio captioning models commonly emit a single sentence regardless of input length [18]. Even modern audio language models with longer context are evaluated on at most five minutes [19, 20].

We argue that the task should mimic human listening: understanding continuous audio as a segmented record, leading to the LOGBOOK benchmark. **Task.** We newly define long-form audio understanding as a gap-free segmentation in which every segment carries an event label and a description (Figure 1). **Benchmark.** We curate publicly available datasets into a unified format, with uninterrupted recordings from ten minutes to six days (Table 1), and design an evaluation suite scoring segmentation and description quality, with a human baseline. **Extensive comparisons.** We compare 48 cascaded systems pairing six audio captioning models with eight text LLMs, and four end-to-end (E2E) audio language models. We also ablate fine-tuning, context window size, and reasoning budget.

¹Supplementary materials, including § A, are available at <https://github.com/facebookresearch/logbook>.

Fig. 1: Example input and output schema

```
Input = {
  "audio": ..., // Raw audio
  "label": { // Class definitions
    "leisure": "Enjoyable activities.",
    "traveling": "Commuting or travel
                 between activities.", ... }
}
Output = [{
  "start": 0, "end": 24, // In minutes
  "event": "traveling",
  "description": "The wearer is
                  commuting by taking the subway."
}, ... // Other actions]
```

We conclude that gap-free segmentation is tractable from audio alone. However, systems often over-segment, while fine-tuning partially mitigates such. Also, E2E are often stronger than cascaded systems in both performance and compute, yet degrade with longer windows. Reasoning yields limited gains, mainly by improving instruction following rather than audio understanding.

2. THE LOGBOOK BENCHMARK

2.1. Task definition (Figure 1)

Input. A continuous recording, ranging from minutes to days, and a prompt specifying the output format and a fixed vocabulary of event labels with their definitions (full prompt in § A.6).

Output. Non-overlapping, contiguous segments that fully cover the recording, each with one event label and a free-text description.

Rationale. The label vocabulary sets the granularity of segmentation. For example, cooking and eating can collapse into a single *food* segment. The description carries the information that the label omits, so that the pair accounts for the full content of the segment. Requiring full coverage also keeps systems comparable: every second is assigned, so a system cannot raise precision by abstaining.

2.2. Datasets

Ego4D [6] is an egocentric video dataset covering daily-life activity across a wide range of scenarios. We use their Summary and Moments tiers. In the Summary tier, two human annotators independently cover each recording, writing one to three sentences per five-minute window. The Moments tier localizes individual actions from a 110-class vocabulary, where most last under a minute.

EgoLife [7] records six participants living together for one week. We use its dense captions: first-person Mandarin, averaging 2.7s per caption. Sessions are formed by cutting audio at gaps over one second and merging caption fragments less than a minute apart.

Table 1. Dataset statistics. # sess. is the number of uninterrupted sessions and # subj. is the number of unique recorded subjects.

Source	Split	# sess.	Avg. len.	Tot. len.	Label	Domain	# subj.
Ego4D	Train	1350	24.7 min	555.8 h	Inferred	Egocentric	250
Ego4D	Val	247	26.1 min	107.6 h	Inferred	Egocentric	71
Ego4D	Test	163	30.2 min	82.1 h	Inferred	Egocentric	55
EgoLife	Test	170	84.9 min	240.6 h	Inferred	Egocentric	6
SINS	Test	1	148.9 h	148.9 h	Provided	Smart home	1

SINS [21] captures six days of a single inhabitant, recorded by twelve microphone arrays throughout a house. It provides contiguous event labels over 17 event classes, but no descriptions.

2.3. Dataset preprocessing

Ego4D. We use OLMo-2-1124-7B-SFT [22] for all text preprocessing. Each five-minute narration slice passes three stages. *Clean* stage formats the raw narration into a third person summary. *Atomic fact generation (AFG)* stage decomposes the cleaned summary into atomic facts, following [23, 24]. *Classify* stage assigns one of six activity classes (productive, food, leisure, purchasing, traveling, other), adapted from the American Time Use Survey [25] (full definition in § A.6). Then, adjacent slices with the same class are merged. Ego4D splits are assigned 60/20/20 by hashing the participant identifier, so the assignment is deterministic and no participant appears in two splits. Ego4D’s recordings are unevenly distributed across capture sites, and we keep only the four sites with longer than ten test set hours: ‘iiith’, ‘minnesota’, ‘kaust’, and ‘frl_track_1_public’. Finally, for the test split, we filter the recordings that are shorter than ten minutes or have fewer than two events.

EgoLife. We follow the similar pipeline to Ego4D. We first translate the dense captions from Mandarin with NLLB-200-distilled-600M [26], then group them into five-minute windows to match Ego4D’s granularity. The label vocabulary for *Classify* stage is reused from Ego4D unchanged, and each caption is directly used as atomic facts. Finally, we apply the same test set filter to Ego4D.

SINS. We create a mono-stream audio using the ground-truth annotation. We select the array nearest the inhabitant by crossfading over two seconds. Unannotated intervals are filled with absence (17% of full audio), making the label timeline fully contiguous.

2.4. Evaluation

Segmentation. *Boundary F1 (bF1)* measures whether boundaries are placed correctly. A boundary counts as matched if it falls within 150s. Harmonic mean is calculated between precision and recall. *Event error rate (evER)* discards length and compares only the predicted label sequence to the reference by the Levenshtein distance, being a diagnostic for over-segmentation. *Frame-level Accuracy (fAcc)* reports the fraction of correctly labelled frames, penalizing boundary and label errors jointly. We use it as the headline metric. bF1 and fAcc are micro-averaged, and evER is macro-averaged.

Description. Similar to FactScore [23, 24], predicted descriptions are decomposed into atomic facts by the same OLMo model, and each fact is compared against the reference facts with BART fine-tuned on text entailment (BART-large-MNLI [27]). Predicted and reference fact pairs within the temporally overlapping segments are compared, and a pair counts as matched when the entailment probability exceeds 0.5. *Recall (dR)* is the fraction of reference facts entailed by some predicted fact, and *Precision (dP)* the fraction of predicted facts entailed by some reference fact, where both are micro-averaged. *F1 (dF1)* is their harmonic mean, the headline metric.

Compute. We report *peta-FLOPs (PFLOPs)* to process the first 24 h of SINS. We first measure PFLOPs of one chunk, multiply by the number of chunks, then add an autoregressive approximation of [28] ($2 \times \#$ of generated tokens $\times \#$ of active parameters).

Downstream utility. We additionally measure two metrics for Ego4D. *Hit rate* is recall@1 against the Moments tier: for each annotated moment, whether any temporally overlapping predicted fact entails it. *Redundancy (rddc)* is the fraction of predicted facts entailed by another fact within the same segment.

Human reference. Ego4D has two annotation passes, so we score each against the other and pool both directions. Note that this is a proxy: the labels were inferred by an LLM and carry its errors; both passes sit on a five-minute grid, earning no partial credit for bF1 under 150s tolerance, unlike minute-resolution systems; annotators vary in detail; and annotators saw video, which systems do not.

3. SYSTEMS

We compare two system designs. End-to-end (E2E) systems consume the audio directly and predict the segments, each with an event label and a description. Cascaded (Casc.) systems first caption short audio windows, then pass the captions to a text LLM to emit the same output format. Detailed prompts are in § A.6.

3.1. End-to-end systems

We use four multimodal models with long context windows for audio. *Qwen 2.5-Omni-7B* [29] (Qwen 2.5-o) reads and generates both text and speech, with strong general audio understanding. *Qwen 3-Omni-30B-A3B-Instruct* [20] (Qwen 3-o) is its successor, a mixture-of-experts model [30] with 30B total and 3B active parameters. *Audio Flamingo 3* [19] (AF 3) targets audio and text only, aligning a pretrained Whisper encoder [31] with a Qwen 2.5-7B decoder [32]. *Gemini 3.5 Flash* [33] (Gemini 3.5) is a closed frontier model that accepts audio natively.

3.2. Cascaded systems

Audio captioning models. The three open E2E models (Qwen 2.5-o, Qwen 3-o, and AF 3) double as captioners. We add Qwen 3-Omni-30B-A3B-Captioner (Qwen 3-o Cap.), the caption-tuned variant of Qwen 3-o, and two CLAP models, *EnCLAP-large* [34] and *MSCLAP-2023* [35] (clapcap version), motivated by the finding that CLAP can outperform larger audio-language models at audio pairwise comparison task [36].

Text models. We test eight instruction-tuned text models. Fully open (data, code, and weights): *OLMo 3.1-32B-Instruct* [37] (OLMo 3.1) and *K2-V2-70B-Instruct* [38] (K2-V2). Open weights, following the SPAN temporal reasoning benchmark [39]: *Qwen 2.5-72B-Instruct* [32] (Qwen 2.5) and *Llama 3.3-70B-Instruct* [40] (Llama 3.3). Frontier open weights, sized comparably to OLMo 3.1: *Qwen 3-32B* [41] (Qwen 3) and *Gemma 4-31B-it* [42] (Gemma 4). Closed APIs, lightweight tier: *Gemini 3.5 Flash* [33] (Gemini 3.5) and *GPT 5.6 Luna* [43] (GPT 5.6).

4. RESULTS

Settings. E2E models receive ten-minute windows, the longest AF 3 accepts. Captioners receive ten-second windows, matching the AudioSet format [3]. The captions are then grouped into ten minutes for the text model, for a fair comparison to E2E. Each window is processed independently, decoded greedily, and parsed leniently [44], except for CLAP models, which use beam search as default. Boundaries are predicted at a minute-level resolution.

Table 2. Comparing audio captioning models across three datasets. Metrics averaged across eight text models in § 3.2. Shaded columns highlight the headline metrics for the segmentation and description tasks. The human row is a proxy rather than a ceiling; see § 2.4.

Model	PFLOPs	Ego4D								SINS			EgoLife					
		Segmentation			Description			Utility		Segmentation			Segmentation			Description		
		bF1↑	evER↓	fAcc↑	dR↑	dP↑	dF1↑	Hit↑	Rddc↓	bF1↑	evER↓	fAcc↑	bF1↑	evER↓	fAcc↑	dR↑	dP↑	dF1↑
EnCLAP	6.41	0.480	2.528	0.320	0.222	0.164	0.188	0.151	0.349	0.444	2.072	0.287	0.409	2.623	0.308	0.080	0.504	0.138
MSCLAP	0.48	0.487	2.618	0.287	0.219	0.156	0.181	0.142	0.308	0.394	2.661	0.395	0.432	2.606	0.302	0.078	0.468	0.132
Qwen 2.5-o	43.01	0.491	3.519	0.295	0.240	0.146	0.181	0.150	0.282	0.432	2.613	0.395	0.450	4.205	0.288	0.093	0.427	0.150
Qwen 3-o	6.69	0.502	4.444	0.313	0.283	0.140	0.186	0.161	0.282	0.358	3.864	0.354	0.460	4.877	0.295	0.114	0.426	0.178
Qwen 3-o Cap.	9.24	0.450	4.499	0.283	0.271	0.136	0.173	0.144	0.294	0.289	4.312	0.255	0.442	5.052	0.277	0.111	0.429	0.172
AF 3	54.48	0.488	2.757	0.330	0.245	0.166	0.197	0.165	0.320	0.468	2.303	0.437	0.437	3.134	0.293	0.088	0.503	0.149
Human	—	0.460	0.745	0.646	0.550	0.550	0.550	0.329	0.227	—	—	—	—	—	—	—	—	—

Table 3. Comparing text models across three datasets. Metrics averaged across six audio captioning models in § 3.2.

Model	PFLOPs	Ego4D								SINS			EgoLife					
		Segmentation			Description			Utility		Segmentation			Segmentation			Description		
		bF1↑	evER↓	fAcc↑	dR↑	dP↑	dF1↑	Hit↑	Rddc↓	bF1↑	evER↓	fAcc↑	bF1↑	evER↓	fAcc↑	dR↑	dP↑	dF1↑
K2-V2	62.11	0.404	1.773	0.117	0.107	0.106	0.101	0.061	0.223	0.244	1.277	0.186	0.322	1.526	0.117	0.035	0.402	0.063
OLMo 3.1	18.77	0.496	3.685	0.179	0.221	0.157	0.184	0.139	0.331	0.332	3.339	0.219	0.498	4.685	0.170	0.100	0.483	0.165
Qwen 2.5	39.60	0.492	3.361	0.367	0.270	0.141	0.183	0.172	0.282	0.423	3.484	0.337	0.436	3.683	0.352	0.099	0.448	0.160
Llama 3.3	41.36	0.500	4.565	0.363	0.272	0.151	0.193	0.162	0.351	0.332	4.688	0.331	0.471	4.724	0.369	0.111	0.457	0.178
Qwen 3	19.70	0.481	3.472	0.308	0.271	0.148	0.190	0.162	0.328	0.387	3.499	0.337	0.451	4.086	0.345	0.107	0.444	0.172
Gemma 4	16.93	0.500	3.511	0.370	0.271	0.189	0.219	0.166	0.293	0.435	3.103	0.442	0.444	3.682	0.326	0.095	0.504	0.157
Gemini 3.5	—	0.492	3.096	0.373	0.271	0.156	0.197	0.175	0.317	0.507	1.927	0.513	0.441	3.337	0.334	0.093	0.467	0.153
GPT 5.6	—	0.499	3.691	0.361	0.289	0.163	0.205	0.180	0.319	0.518	2.450	0.467	0.446	4.276	0.336	0.112	0.473	0.179
Human	—	0.460	0.745	0.646	0.550	0.550	0.550	0.329	0.227	—	—	—	—	—	—	—	—	—

4.1. Comparing different components of cascaded systems

A cascaded system pairs one captioner with one text model, total of $6 \times 8 = 48$ systems. Table 2 averages over the text models for each captioner, and vice versa for Table 3. Full results are in § A.7.

Among captioners in Table 2, AF 3 leads on Ego4D and SINS, though it is also the most computationally heavy. MSCLAP is not the best, but comes close to AF 3 at a fraction of the cost. Within the Qwen family, Qwen 3-o is best everywhere except SINS, and interestingly, beats its own captioner-tuned variant throughout.

Among text models (Table 3), closed APIs are often the best, where Gemma 4 is generally the strongest non-API model. Additionally, K2-V2’s low evER is an artifact of returning no parseable output for 64% of the windows. As such, for all the below experiments, we use AF 3 as the captioner, and compare three text models, Gemini 3.5, GPT 5.6, and Gemma 4, for the cascaded systems.

4.2. Comparing cascaded systems with E2E systems

We now compare the three best cascaded systems from § 4.1 with four E2E systems in Table 4 on headline metrics, fAcc and dF1. E2E Gemini 3.5 is the best on Ego4D and EgoLife across both systems, beating cascaded Gemini 3.5 on both. SINS is the exception, as the three cascade systems lead the four E2E systems. We suspect this might be related to SINS being longer (Table 1); § 5.2 examines how systems respond to longer input windows.

Among non-API E2E models, Qwen 3-o is the strongest, while AF 3, the best captioner, scores zero as it fails to emit the required output schema. Also, Qwen 2.5-o’s high EgoLife fAcc is due to the model predicting *productive* almost everywhere, and 64.3% of EgoLife frames are *productive*. We caution that fAcc rewards the majority class, so we additionally report class-wise accuracy in § A.8.

Finally, we emphasize that cascaded systems are not the cheaper option. Even counting only the text model, Gemma 4 costs 16.93 PFLOPs against 11.97 for E2E Qwen 3-o, and adding the AF 3 captioner brings the cascade to 71.41.

5. ANALYSIS

Both cascaded and E2E systems in § 4 fall short of the human baseline on Ego4D. In particular, even the best systems suffer from the **over-segmentation problem** (Figure 2), showing evER above 100%, with GPT 5.6 reaching 286% and 343% on Ego4D and EgoLife, respectively. We test three additional methods that may improve the performance: supervised fine-tuning (§ 5.1), longer context (§ 5.2), and reasoning (§ 5.3)

5.1. Supervised fine-tuning (SFT)

As SFT may teach the model both the output format and how finely to segment, we fine-tune Gemma 4 and Qwen 3-o, which are the strongest non-API systems for the cascaded and E2E systems of § 4.2, on the Ego4D train split, with frozen AF 3 captions as Gemma 4’s input. We employ LoRA [45] ($r=16$, $\alpha=32$, dropout=0.05) for one epoch of training, and use AdamW [46] at batch size 8 and learning rate $1e-4$ with a cosine schedule and 3% warmup, best-kept by valid loss every 200 steps.

Indeed, Table 4 demonstrates that SFT mitigates the over-segmentation problem: evER falls below 100% for both cascaded Gemma 4 and E2E Qwen 3-o on all datasets, and fAcc sharply improves on Ego4D. Figure 2 further shows that the predicted segmentations become coarser, with the event count falling and mean event length rising on every dataset (average event counts in § A.4).

However, for SINS and EgoLife, which are out of distribution, the gains are much smaller, and Gemma 4 even degrades on SINS. One reason is class coverage: SINS consists of 27% *sleeping* and 17% *absence*, classes that fine-tuned models avoid because Ego4D contains only events with actions (label distribution in § A.8).

Also, EgoLife description distribution differs with Ego4D. Ego4D averages 3.6 facts per five-minute window against EgoLife’s 48.7 (additional statistics in § A.3). SFT therefore teaches the model to write far less, leading to an increase in dP but a decrease in dR.

Table 4. Comparing cascaded and E2E systems across three datasets. Cascaded systems use AF 3 as the audio captioning model. Shaded columns highlight the headline metrics for the segmentation and description tasks. The human row is a proxy rather than a ceiling; see § 2.4.

System	PFLOPs	Ego4D								SINS			EgoLife					
		Segmentation			Description			Utility		Segmentation			Segmentation			Description		
		bF1↑	evER↓	fAcc↑	dR↑	dP↑	dF1↑	Hit↑	Rddc↓	bF1↑	evER↓	fAcc↑	bF1↑	evER↓	fAcc↑	dR↑	dP↑	dF1↑
Casc. Gemini 3.5	—	0.493	2.319	0.397	0.276	0.175	0.215	0.198	0.340	0.612	1.133	0.571	0.429	2.686	0.325	0.087	0.512	0.149
Casc. GPT 5.6	—	0.500	2.863	0.383	0.273	0.182	0.218	0.191	0.340	0.640	1.284	0.527	0.428	3.429	0.327	0.096	0.510	0.162
Casc. Gemma 4	71.41	0.508	2.608	0.398	0.262	0.218	0.238	0.160	0.304	0.547	1.588	0.512	0.430	2.822	0.311	0.085	0.560	0.147
+ SFT	71.12	0.247	0.785	0.520	0.316	0.403	0.355	0.180	0.207	0.573	0.668	0.421	0.254	0.756	0.335	0.044	0.706	0.084
E2E Gemini 3.5	—	0.437	1.357	0.445	0.271	0.209	0.236	0.214	0.289	0.368	2.256	0.495	0.422	1.648	0.331	0.087	0.501	0.149
E2E Qwen 3-o	11.97	0.349	1.344	0.413	0.214	0.161	0.183	0.191	0.347	0.415	1.950	0.426	0.334	1.469	0.273	0.082	0.463	0.139
+ SFT	11.67	0.252	0.681	0.633	0.376	0.481	0.422	0.254	0.212	0.285	0.721	0.483	0.241	0.770	0.332	0.057	0.773	0.107
E2E Qwen 2.5-o	22.71	0.359	1.404	0.398	0.100	0.306	0.151	0.131	0.389	0.289	2.980	0.275	0.349	0.913	0.638	0.056	0.643	0.103
E2E AF 3	51.51	0.000	1.000	0.000	0.000	—	—	0.000	0.000	0.000	1.000	0.000	0.000	1.000	0.000	0.000	—	—
Human	—	0.460	0.745	0.646	0.550	0.550	0.550	0.329	0.227	—	—	—	—	—	—	—	—	—

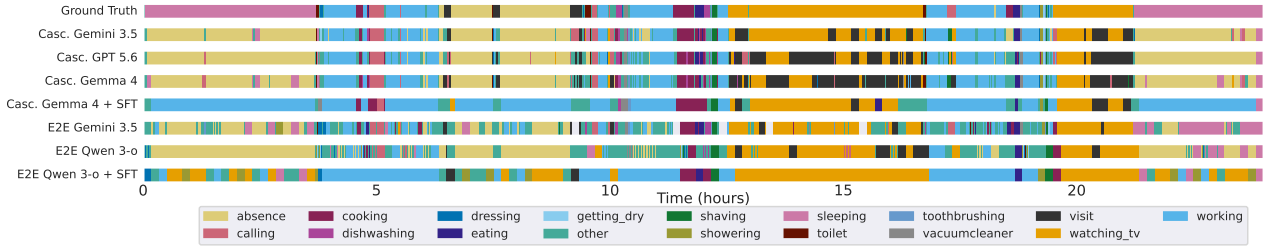


Fig. 2. Visualization of the predicted segmentations for the first 24 hours of SINS dataset using SINS label definition.

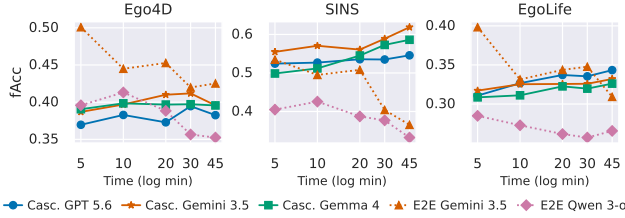


Fig. 3. Segmentation performance ablation on context window size.

Finally, comparing the two best SFT systems, E2E Qwen 3-o improves more: bigger gains on Ego4D and EgoLife, and it alone gains on SINS. The two gains differ in character. Gemma 4 benefits more from learning granularity, reducing its Ego4D event count $4.7\times$ against $2.1\times$ for Qwen 3-o (full event count statistics in § A.4). Qwen 3-o instead benefits more from better labeling: its *productive* fAcc rises from 27% to 76%, against 32% to 60% for Gemma 4.

5.2. Context window size

To test whether longer context helps, we vary the window from 5 to 45 minutes for the top-performing systems of § 4.2. Figure 3 shows that segmentation quality for cascaded systems is similar or better at longer windows, whereas for E2E systems it peaks at 5 or 10 minutes and then declines. The results suggest that cascaded systems can exploit longer context better than E2E systems. Additionally, description precision tends to rise while recall falls, as predicted length of each segment becomes longer (average length statistics in § A.1).

5.3. Reasoning

We enable reasoning capabilities of the topline models. Following [47], we use hard budget caps of 1024, 2048, and 4096 tokens for Qwen 3-o and Gemma 4, where we use the Thinker variant for Qwen 3-o. We use the same values as Gemini’s soft cap, and low, medium, and high reasoning effort for GPT 5.6.

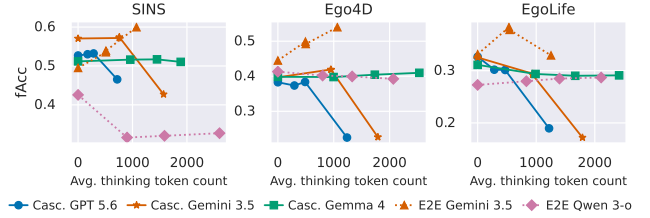


Fig. 4. Segmentation performance ablation on reasoning budget.

As shown in Figure 4, only E2E Gemini benefits, its segmentation performance rises monotonically with budget. Every other system is similar or worse. Much of this is explained by the models’ failure to emit the required output format. E2E Gemini’s gain coincides with correcting its parsing failures from 15.4% to nearly none (additional parsing failure statistics in § A.5). The drops for cascaded Gemini 3.5 and GPT 5.6 at the highest budget come from output truncation, as reasoning exhausts the output budget. Cascaded Gemma 4 and E2E Qwen 3-o already parse well, and more thinking slightly worsens or does not change it, leading to no gains. The trend is the same for all the other datasets. The results suggest reasoning helps instruction following rather than audio understanding.

6. CONCLUSION

We introduce the LOGBOOK benchmark, designing the task of long-form audio understanding as a gap-free segmentation, where each segment carries an event label and a description. We demonstrate that the task, even without vision, is feasible with currently available models. However, all systems over-segment, which SFT mitigates. Also, we show that E2E can outperform cascaded systems on both headline metrics and efficiency, yet degrade with longer windows. Finally, we show that reasoning yields limited gains by improving instruction following rather than audio understanding. Future works can be done on additional event vocabularies, structured decoding, reinforcement learning, long context for E2E models, and assistive uses for deaf and hard-of-hearing users or personalized assistants.

7. COMPLIANCE WITH ETHICAL STANDARDS

This study relied entirely on publicly available datasets and models. Consequently, ethical approval was not required.

8. ACKNOWLEDGMENTS

We thank Ray Huang and Gautam Tiwari for valuable discussions.

9. REFERENCES

- [1] Zhaojiang Lin et al., “Wearvox: An egocentric multichannel voice assistant benchmark for wearables,” in *ICLR*, 2026.
- [2] Akshay Paruchuri et al., “Egotrigger: Toward audio-driven image capture for human memory enhancement in all-day energy-efficient smart glasses,” *IEEE TVCG*, 2025.
- [3] Jort F Gemmeke et al., “Audio set: An ontology and human-labeled dataset for audio events,” in *ICASSP*, 2017.
- [4] Romain Serizel et al., “Sound event detection in synthetic domestic environments,” in *ICASSP*, 2020.
- [5] Konstantinos Drossos et al., “Clotho: an audio captioning dataset,” in *ICASSP*, 2020.
- [6] Kristen Grauman et al., “Ego4d: Around the world in 3,000 hours of egocentric video,” *IEEE TPAMI*, 2024.
- [7] Jingkan Yang et al., “Egolife: Towards egocentric life assistant,” in *CVPR*, 2025.
- [8] Kris M Kitani et al., “Fast unsupervised ego-action learning for first-person sports videos,” in *CVPR*, 2011.
- [9] Omid Aghazadeh et al., “Novelty detection from an ego-centric perspective,” in *CVPR*, 2011.
- [10] Junhao Pan et al., “Youhome system and dataset: Making your home know you better,” *IEEE iSES*, 2022.
- [11] Luca Fredianelli et al., “Environmental noise dataset for sound event classification and detection,” *Scientific Data*, 2025.
- [12] Dima Damen et al., “The epic-kitchens dataset: Collection, challenges and baselines,” *IEEE TPAMI*, 2020.
- [13] Toby Perrett et al., “HD-EPIC: A highly-detailed egocentric video dataset,” in *CVPR*, 2025.
- [14] Orevaoghene Ahia et al., “Blab: Brutally long audio bench,” *arXiv preprint 2505.03054*, 2025.
- [15] Kaiwen Luo et al., “Chronosaudio: A comprehensive long-audio benchmark for evaluating audio-large language models,” *arXiv preprint 2601.04876*, 2026.
- [16] Mingchen Shao et al., “Listening with time: Precise temporal awareness for long-form audio understanding,” *arXiv preprint 2604.22245*, 2026.
- [17] Annamaria Mesaros et al., “Sound event detection: A tutorial,” *IEEE SPM*, 2021.
- [18] Xinhao Mei et al., “Automated audio captioning: An overview of recent progress and new challenges,” *Journal on Audio, Speech, and Music Processing*, 2022.
- [19] Sreyan Ghosh et al., “Audio flamingo 3: Advancing audio intelligence with fully open large audio language models,” *NeurIPS*, 2026.
- [20] Jin Xu et al., “Qwen3-omni technical report,” *arXiv preprint 2509.17765*, 2025.
- [21] Gert Dekkers et al., “The sins database for detection of daily activities in a home environment using an acoustic sensor network,” *DCASE*, 2017.
- [22] Evan Pete Walsh et al., “2 olmo 2 furious,” in *COLM*, 2025.
- [23] Sewon Min et al., “FactScore: Fine-grained atomic evaluation of factual precision in long form text generation,” in *EMNLP*, 2023.
- [24] Lucas Fonseca Lage and Simon Ostermann, “Openfactscore: Open-source atomic evaluation of factuality in text generation,” *arXiv preprint 2507.05965*, 2025.
- [25] U.S. Bureau of Labor Statistics, “American time use survey,” Tech. Rep., U.S. Bureau of Labor Statistics, June 2026.
- [26] Marta R Costa-Jussà et al., “No language left behind: Scaling human-centered machine translation,” *arXiv preprint 2207.04672*, 2022.
- [27] Mike Lewis et al., “Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension,” in *ACL*, 2020.
- [28] Jared Kaplan et al., “Scaling laws for neural language models,” *arXiv preprint 2001.08361*, 2020.
- [29] Jin Xu et al., “Qwen2.5-omni technical report,” *arXiv preprint 2503.20215*, 2025.
- [30] Noam Shazeer et al., “Outrageously large neural networks: The sparsely-gated mixture-of-experts layer,” in *ICLR*, 2017.
- [31] Alec Radford et al., “Robust speech recognition via large-scale weak supervision,” in *ICML*, 2023.
- [32] An Yang et al., “Qwen2.5 technical report,” *arXiv preprint 2412.15115*, 2025.
- [33] Google, “Gemini 3.5 flash,” <https://ai.google.dev/gemini-api/docs/models/gemini-3.5-flash>, 2026.
- [34] Jaeyeon Kim et al., “Enclap: Combining neural audio codec and audio-text joint embedding for automated audio captioning,” in *ICASSP*, 2024.
- [35] Benjamin Elizalde et al., “Natural language supervision for general-purpose audio representations,” in *ICASSP*, 2024.
- [36] Tianyu Guo et al., “Brace: A benchmark for robust audio caption quality evaluation,” *NeurIPS*, 2026.
- [37] Team Olmo et al., “Olmo 3,” *arXiv preprint 2512.13961*, 2025.
- [38] Zhengzhong Liu et al., “K2-v2: A 360-open, reasoning-enhanced llm,” *arXiv preprint 2512.06201*, 2025.
- [39] Zhongjian Miao et al., “Span: Benchmarking and improving cross-calendar temporal reasoning of large language models,” in *AAAI*, 2026.
- [40] Aaron Grattafiori et al., “The llama 3 herd of models,” *arXiv preprint 2407.21783*, 2024.
- [41] An Yang et al., “Qwen3 technical report,” *arXiv preprint 2505.09388*, 2025.
- [42] Gemma Team et al., “Gemma 4 technical report,” *arXiv preprint 2607.02770*, 2026.
- [43] OpenAI, “Gpt-5.6 luna,” <https://developers.openai.com/api/docs/models/gpt-5.6-luna>, 2026.
- [44] Stefano Baccianella, “Json repair: A python module to repair invalid json,” feb 2025.
- [45] Edward J Hu et al., “Lora: Low-rank adaptation of large language models,” in *ICLR*, 2022.
- [46] Ilya Loshchilov and Frank Hutter, “Decoupled weight decay regularization,” in *ICLR*, 2019.
- [47] Niklas Muennighoff et al., “s1: Simple test-time scaling,” in *EMNLP*, 2025.

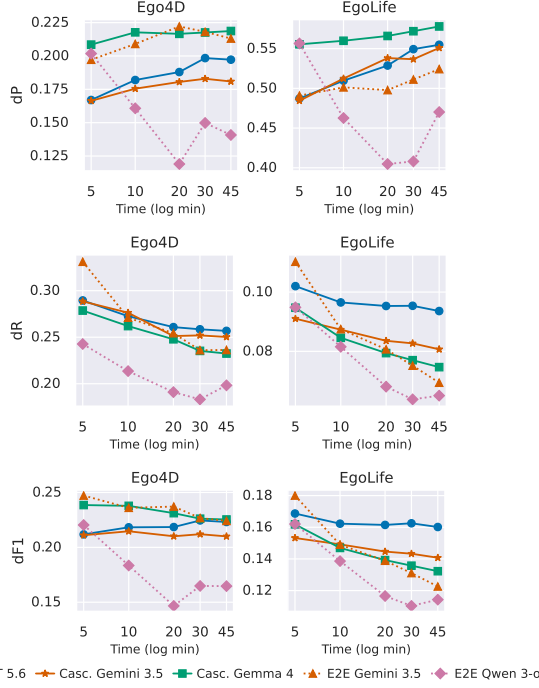


Fig. 5. Description performance ablation on the context window size.

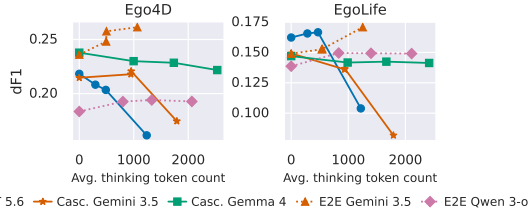


Fig. 6. Description performance ablation on the reasoning budget.

A. APPENDIX

A.1. Ablation on context window size for description quality

Table 5 reports the average number of predicted events per five minutes and their average length. Longer input windows produce longer predicted events across every system, and correspondingly fewer facts. Figure 5 shows the effect on description quality: precision generally rises while recall falls, and the fall in recall dominates, so dF1 decreases overall.

A.2. Ablation on reasoning budget for description quality

Figure 6 reports description quality with respect to reasoning budget. It follows nearly the same trend as segmentation in Figure 4: E2E Gemini 3.5 improves, cascaded Gemini 3.5 and GPT 5.6 collapse at the largest budget, and the rest are flat. Since a window that fails to parse yields neither segments nor facts, this consistency further supports attributing the changes to parsing failures rather than to improved understanding.

Table 5. Events per 5-minute interval across Ego4D, EgoLife, and SINS datasets. Average counts per five minutes, and average length for each event in seconds.

System	Window	Ego4D		EgoLife		SINS	
		Cnt.	Len.	Cnt.	Len.	Cnt.	Len.
Casc. Gemini 3.5	5min	1.84	162	1.57	191	0.88	341
	10min	1.26	236	1.12	267	0.57	525
	20min	1.08	275	0.90	330	0.43	695
	30min	1.05	282	0.90	332	0.43	697
	45min	1.05	283	0.90	328	0.46	659
Casc. GPT 5.6	5min	1.89	158	1.58	190	0.87	343
	10min	1.51	195	1.31	223	0.63	474
	20min	1.40	211	1.12	268	0.51	588
	30min	1.31	228	1.10	272	0.43	705
	45min	1.26	232	0.97	309	0.44	677
Casc. Gemma 4	5min	1.83	164	1.53	196	1.04	289
	10min	1.40	214	1.17	256	0.73	412
	20min	1.25	240	1.03	291	0.66	455
	30min	1.08	274	0.95	315	0.63	478
	45min	1.08	276	0.83	359	0.59	509
E2E Gemini 3.5	5min	0.94	305	1.02	281	1.38	216
	10min	0.66	384	0.64	384	0.95	307
	20min	0.54	466	0.54	464	0.79	368
	30min	0.49	506	0.47	517	0.75	367
	45min	0.46	543	0.39	602	0.66	418
E2E Qwen 3-o	5min	0.92	324	0.91	328	1.46	205
	10min	0.60	500	0.49	606	0.83	360
	20min	0.56	533	0.70	425	1.04	289
	30min	0.47	632	0.72	415	0.55	538
	45min	0.25	1175	0.23	1296	0.16	1797
GT	–	0.57	528	0.40	754	0.32	952

A.3. Fact statistics

Table 6 reports the average number of atomic facts per five-minute interval. Where every system produces a similar amount of facts on both datasets, the ground truth differing more than thirteenfold (3.64 on Ego4D against 48.67 on EgoLife). SFT roughly halves it, bringing it close to the Ego4D reference it was trained on and further from EgoLife’s.

A.4. Event Statistics

Table 7 reports predicted event counts and lengths before and after SFT. GT is calculated only with the first annotator for Ego4D. SFT cuts the Ego4D event count $4.7\times$ for the cascade and $2.1\times$ for the E2E system, in both cases past the reference: both settle near 0.3 events per five minutes, with the same pattern on EgoLife and SINS.

A.5. Parsing Failure Statistics

Table 8 reports the fraction of output predictions whose output could not be parsed into the required schema. Two model-side failures that often arise: outputs that terminate normally but do not follow the schema, or outputs truncated when the budget is exhausted. At baseline, E2E Gemini’s outputs terminate normally yet do not follow the schema; reasoning corrects this and its failure rate falls from 15.4% to 0.6% on Ego4D. At the largest budget, cascaded Gemini and GPT 5.6 instead exhaust their completion budget on reasoning tokens and their outputs are truncated, pushing failures to 49.2% and 43.1%.

Table 6. Facts per 5-minute interval for Ego4D and EgoLife datasets. We use first pass annotations for Ego4D to count the facts.

System	Ego4D	EgoLife
Casc. Gemini 3.5	9.17	9.21
Casc. GPT 5.6	18.75	18.94
Casc. Gemma 4	9.89	10.06
+ SFT	4.62	4.91
E2E Gemini 3.5	8.50	10.10
E2E Qwen 3-o	9.72	10.61
+ SFT	4.62	5.11
GT	3.64	48.67

Table 7. Events per 5-minute interval across Ego4D, EgoLife, and SINS datasets. Average counts per five minutes, and average length for each event in seconds.

System	Ego4D		EgoLife		SINS	
	Cnt.	Len.	Cnt.	Len.	Cnt.	Len.
Casc. Gemini 3.5	1.26	236	1.12	267	0.57	525
Casc. GPT 5.6	1.51	195	1.31	223	0.63	474
Casc. Gemma 4	1.40	214	1.17	256	0.73	412
+ SFT	0.30	995	0.23	1296	0.21	1403
E2E Gemini 3.5	0.66	384	0.64	384	0.95	307
E2E Qwen 3-o	0.60	500	0.49	606	0.83	360
+ SFT	0.28	1051	0.24	1263	0.22	1364
GT	0.57	528	0.40	754	0.32	952

Table 8. Fraction of predictions not following the required format.

System	Budget	Ego4D	EgoLife	SINS
Casc. Gemini 3.5	None	0.6%	0.2%	0.0%
	1024	1.7%	3.0%	0.9%
	2048	2.4%	2.8%	0.9%
	4096	49.2%	48.2%	31.1%
Casc. GPT 5.6	None	0.5%	0.1%	0.0%
	Low	1.3%	0.9%	0.4%
	Medium	1.1%	0.4%	0.1%
	High	43.1%	38.8%	16.1%
Casc. Gemma 4	None	0.4%	0.2%	0.0%
	1024	2.1%	1.5%	0.7%
	2048	1.2%	0.5%	0.4%
	4096	0.5%	0.3%	0.3%
E2E Gemini 3.5	None	15.4%	17.4%	3.4%
	1024	7.4%	7.7%	2.0%
	2048	7.5%	7.4%	2.2%
	4096	0.6%	0.2%	0.2%
E2E Qwen 3-o	None	0.3%	0.2%	0.2%
	1024	0.5%	0.3%	0.1%
	2048	0.3%	0.2%	0.2%
	4096	0.3%	0.1%	0.1%

A.6. Prompts and Class definitions

Fig. 7: Prompt for E2E systems

You are an expert audio analyst for wearable-device recordings of a single person’s activities. You receive long audio recordings, to produce a temporal segmentation of the wearer’s activities.

Task

The recording is 10 minutes long.

Segment it into non-overlapping, contiguous spans at *whole-minute* boundaries, covering every minute from 0 to 10 using labels from the list below (no gaps). Each segment must span at least one full minute; do not emit sub-minute segments.

Output format

Return a JSON object with one key, "segments", whose value is an array of objects each with:

- "start": integer minute (0..10) * relative to the start of this chunk* ("0" = chunk start)
- "end": integer minute (0..10) * relative to the start of this chunk*
- "event": one of the allowed event classes below
- "description": 1-2 sentences, present tense, naming the concrete actions and objects in this segment. Add detail beyond the event label; infer plausible specifics when evidence is partial. Refer to the recording subject as "the wearer".

Each end must be strictly greater than its start. Spans must be non-overlapping and chronologically ordered. Do not output anything outside the JSON object.

Allowed event classes (use these labels exactly):
(Different per dataset)

Fig. 8: Prompt for Captioners

Describe the audio clip in one sentence

Fig. 9: Prompt for Text LLMs

You are an expert audio analyst for wearable-device recordings of a single person's activities. You receive a series of timestamped descriptions, to produce a temporal segmentation of the wearer's activities.

Task
(Same with E2E)

Output format
(Same with E2E)

Allowed event classes (use these labels exactly):
(Same with E2E)

Audio descriptions
(Output of audio captioners)

Fig. 10: Class definition of Ego4D and EgoLife

- **productive:** Directed activity that produces output or meets someone's tangible needs: paid or unpaid work (including skilled trades and manual labor as a job), studying or attending class, homework or research, household maintenance (cleaning, laundry, repairs, lawn/garden, pet care), and hands-on dependent care (feeding, dressing, bathing, teaching children or elderly). Exclude: cooking or eating (use 'food'); shopping (use 'purchasing'); commuting (use 'traveling'); recreational or conversational time with family (use 'leisure').
- **food:** Preparing, eating, or drinking, alone or with others, at home or elsewhere. Includes cooking, setting the table, and kitchen cleanup. Eating in any context belongs here, not in leisure.
- **leisure:** Activities done for personal enjoyment or relaxation rather than work or output: TV / streaming / movies, music, video games, reading or writing for pleasure, hobbies, arts and crafts, cultural events; exercise and sports of all kinds; personal phone, text, or video calls; in-person socializing or hosting; and casual family time (chatting, playing, reading together). Exclude: household chores or maintenance (cleaning, laundry, repairs, lawn/garden, pet care) and hands-on dependent care (use 'productive'); shopping or in-person transactions (use 'purchasing'); eating with others (use 'food').
- **purchasing:** Shopping for goods or services, in-store, online, or by phone, including visits to professionals such as doctors, banks, or hairdressers.
- **traveling:** Commuting or travel between activities, by car, foot, transit, or any other means. Movement during another activity (e.g., walking around a workshop while working) belongs to that activity.
- **other:** Any other activities that do not fit the above categories. Also use this when there is no audible activity (silence, ambient quiet, or audio where no clear human activity is occurring).

Fig. 11: Class definition of SINS

- absence: nobody present in the home or in the relevant room; ambient room sound only (HVAC, distant traffic, refrigerator hum).
- calling: phone call - sustained one-sided speech with pauses for the remote party, typically in the living room.
- cooking: kitchen sounds: chopping, frying, pot/pan clatter, microwave, sizzling.
- dishwashing: running water in the sink with dishes clinking and scrubbing - manual washing-up, not a dishwasher cycle.
- dressing: fabric rustling, zippers, drawer/wardrobe sounds, hangers - typically in the bedroom.
- eating: cutlery on plates, chewing, glass clinks, brief speech at the table.
- getting_dry: towel rubbing and brisk fabric-on-skin sounds after bathing.
- other: a person IS present and active but the activity doesn't fit any of the listed categories. Includes transitions between activities, time between entering a room and starting something, sitting on the sofa, or crossing through the hall.
- shaving: electric-razor buzz or wet-shave scraping at the sink.
- showering: continuous water spray, often with bathroom-fan hum.
- sleeping: a person IS present but inactive: very quiet room, slow breathing, occasional rustling. Usually a long, uninterrupted stretch.
- toilet: toilet flush, brief water and seat sounds.
- toothbrushing: rhythmic brushing sound, often with a sink running briefly before or after.
- vacuumcleaner: continuous vacuum motor hum, often moving across the room. Same label regardless of which room.
- visit: multiple voices, conversation, doorbell, footsteps - guests in the home.
- watching_tv: TV speech, music, sound effects; periodic dialogue and music cues, often sustained.
- working: working at a computer: keyboard typing, mouse clicks, occasional brief speech (phone or video calls).

A.7. Full results of cascaded systems

Table 9 reports all 48 cascaded systems, *i.e.*, six captioners paired with eight text models, on Ego4D, SINS, and EgoLife. Table 2 and Table 3 average this grid along each axis.

A.8. Class-wise frame accuracy of top-line systems

Table 10 reports the class-wise fAcc for each class, ordered in frequency, *i.e.*, ratio of the event on the full audio. Class-wise fAcc is the fraction of frames of that class the system labels correctly.

Table 9. Comparing all the combinations of cascaded systems.

Audio Cap.	Text model	Ego4D								SINS			EgoLife					
		Segmentation			Description			Utility		Segmentation			Segmentation			Description		
		bF1↑	evER↓	fAcc↑	R↑	P↑	F1↑	Hit↑	rdc↓	bF1↑	evER↓	fAcc↑	bF1↑	evER↓	fAcc↑	R↑	P↑	F1↑
EnCLAP	K2-V2	0.445	1.796	0.189	0.156	0.120	0.136	0.087	0.277	0.301	1.364	0.243	0.372	1.829	0.219	0.059	0.482	0.105
	OLMo 3.1	0.491	2.919	0.177	0.189	0.150	0.168	0.152	0.344	0.361	2.863	0.172	0.496	3.902	0.212	0.089	0.510	0.151
	Qwen 2.5	0.481	2.375	0.374	0.238	0.163	0.194	0.176	0.337	0.497	2.229	0.185	0.381	2.354	0.330	0.078	0.518	0.135
	Llama 3.3	0.501	3.830	0.366	0.247	0.149	0.186	0.163	0.383	0.361	3.613	0.247	0.438	3.040	0.376	0.096	0.465	0.160
	Qwen 3	0.472	2.779	0.294	0.240	0.164	0.195	0.152	0.373	0.409	2.487	0.271	0.399	2.923	0.347	0.091	0.480	0.153
	Gemma 4	0.491	2.184	0.381	0.225	0.207	0.215	0.158	0.336	0.517	1.464	0.310	0.393	2.151	0.317	0.072	0.567	0.127
	Gemini 3.5	0.484	2.039	0.398	0.242	0.176	0.204	0.154	0.369	0.547	1.204	0.479	0.401	2.197	0.329	0.076	0.520	0.133
	GPT 5.6	0.474	2.302	0.384	0.241	0.178	0.205	0.162	0.370	0.559	1.350	0.386	0.390	2.587	0.335	0.081	0.495	0.139
MSCLAP	K2-V2	0.517	1.824	0.126	0.122	0.121	0.122	0.058	0.229	0.219	1.579	0.220	0.402	1.976	0.129	0.035	0.432	0.065
	OLMo 3.1	0.458	2.356	0.139	0.191	0.137	0.160	0.136	0.301	0.308	3.091	0.198	0.508	2.761	0.124	0.085	0.459	0.144
	Qwen 2.5	0.477	2.581	0.355	0.232	0.155	0.186	0.154	0.289	0.419	3.217	0.400	0.389	2.419	0.342	0.082	0.471	0.139
	Llama 3.3	0.515	4.387	0.348	0.260	0.156	0.194	0.147	0.366	0.313	4.984	0.336	0.458	3.766	0.383	0.100	0.452	0.164
	Qwen 3	0.446	2.576	0.229	0.241	0.167	0.197	0.156	0.323	0.366	3.277	0.402	0.446	2.256	0.288	0.089	0.459	0.149
	Gemma 4	0.488	2.422	0.377	0.226	0.194	0.209	0.160	0.308	0.459	2.043	0.563	0.419	2.443	0.374	0.072	0.526	0.126
	Gemini 3.5	0.494	2.240	0.386	0.246	0.153	0.189	0.163	0.325	0.514	1.483	0.521	0.422	2.406	0.415	0.078	0.478	0.134
	GPT 5.6	0.502	2.558	0.340	0.233	0.163	0.192	0.163	0.319	0.552	1.615	0.517	0.416	2.821	0.361	0.080	0.467	0.137
Qwen 2.5-o	K2-V2	0.436	1.800	0.094	0.105	0.099	0.102	0.064	0.202	0.319	1.249	0.210	0.329	1.248	0.062	0.022	0.352	0.041
	OLMo 3.1	0.510	3.595	0.155	0.205	0.142	0.168	0.136	0.281	0.352	3.224	0.249	0.506	5.170	0.150	0.096	0.445	0.159
	Qwen 2.5	0.512	3.393	0.369	0.258	0.138	0.180	0.167	0.254	0.449	3.151	0.378	0.455	4.197	0.374	0.097	0.410	0.156
	Llama 3.3	0.495	4.727	0.351	0.258	0.142	0.184	0.162	0.334	0.344	4.112	0.352	0.473	5.962	0.367	0.111	0.417	0.175
	Qwen 3	0.496	3.675	0.325	0.269	0.134	0.179	0.149	0.297	0.438	2.957	0.364	0.475	5.050	0.360	0.106	0.396	0.167
	Gemma 4	0.493	3.865	0.352	0.270	0.190	0.223	0.158	0.273	0.420	2.812	0.513	0.459	4.034	0.322	0.099	0.478	0.165
	Gemini 3.5	0.492	2.978	0.368	0.263	0.149	0.190	0.181	0.303	0.587	1.293	0.567	0.446	3.062	0.325	0.092	0.437	0.152
	GPT 5.6	0.492	4.122	0.348	0.290	0.176	0.219	0.187	0.315	0.552	2.107	0.529	0.456	4.921	0.341	0.118	0.481	0.189
Qwen 3-o	K2-V2	0.489	2.103	0.108	0.106	0.091	0.098	0.054	0.178	0.290	1.107	0.123	0.330	1.507	0.106	0.037	0.317	0.066
	OLMo 3.1	0.521	5.275	0.230	0.269	0.177	0.214	0.147	0.350	0.339	3.980	0.275	0.501	6.604	0.216	0.120	0.501	0.194
	Qwen 2.5	0.502	4.486	0.364	0.315	0.120	0.173	0.200	0.252	0.367	5.021	0.362	0.477	4.863	0.364	0.124	0.410	0.190
	Llama 3.3	0.490	5.161	0.373	0.301	0.145	0.196	0.152	0.324	0.346	4.721	0.351	0.480	6.173	0.371	0.131	0.436	0.202
	Qwen 3	0.504	4.564	0.334	0.317	0.135	0.189	0.189	0.306	0.357	4.432	0.345	0.486	5.786	0.357	0.133	0.427	0.203
	Gemma 4	0.511	4.845	0.370	0.315	0.166	0.217	0.183	0.271	0.360	4.593	0.442	0.471	4.713	0.317	0.116	0.448	0.185
	Gemini 3.5	0.486	3.975	0.362	0.292	0.151	0.199	0.163	0.297	0.394	2.824	0.498	0.452	3.883	0.308	0.103	0.434	0.167
	GPT 5.6	0.512	5.143	0.363	0.347	0.139	0.198	0.196	0.278	0.407	4.236	0.434	0.483	5.484	0.319	0.143	0.438	0.216
Qwen 3-o Cap.	K2-V2	0.085	1.052	0.007	0.008	0.077	0.015	0.000	0.214	0.031	0.808	0.010	0.057	0.995	0.007	0.002	0.353	0.004
	OLMo 3.1	0.513	4.923	0.177	0.263	0.187	0.218	0.125	0.369	0.274	3.551	0.135	0.506	5.325	0.131	0.113	0.488	0.183
	Qwen 2.5	0.494	4.718	0.360	0.322	0.115	0.169	0.151	0.263	0.345	4.615	0.258	0.492	5.414	0.390	0.127	0.372	0.189
	Llama 3.3	0.495	5.407	0.354	0.294	0.151	0.200	0.160	0.332	0.279	6.314	0.240	0.522	5.082	0.367	0.124	0.479	0.197
	Qwen 3	0.486	4.573	0.341	0.296	0.130	0.181	0.165	0.342	0.291	5.355	0.238	0.477	5.472	0.370	0.126	0.434	0.196
	Gemma 4	0.507	5.141	0.343	0.327	0.160	0.215	0.178	0.264	0.307	6.117	0.311	0.489	5.929	0.311	0.124	0.443	0.194
	Gemini 3.5	0.503	5.023	0.329	0.308	0.133	0.186	0.192	0.270	0.389	3.622	0.440	0.495	5.785	0.305	0.119	0.420	0.185
	GPT 5.6	0.515	5.158	0.349	0.352	0.139	0.199	0.181	0.294	0.399	4.110	0.408	0.500	6.411	0.335	0.155	0.446	0.231
AF 3	K2-V2	0.450	2.066	0.180	0.142	0.129	0.135	0.102	0.239	0.304	1.554	0.308	0.445	1.597	0.182	0.053	0.476	0.096
	OLMo 3.1	0.481	3.044	0.196	0.212	0.151	0.176	0.136	0.340	0.360	3.327	0.284	0.469	4.346	0.185	0.094	0.492	0.158
	Qwen 2.5	0.484	2.614	0.378	0.256	0.155	0.193	0.187	0.297	0.465	2.673	0.436	0.420	2.848	0.309	0.086	0.508	0.148
	Llama 3.3	0.502	3.877	0.388	0.272	0.160	0.201	0.185	0.368	0.352	4.382	0.458	0.452	4.317	0.353	0.102	0.496	0.170
	Qwen 3	0.484	2.667	0.322	0.265	0.156	0.196	0.160	0.330	0.464	2.483	0.402	0.421	3.026	0.351	0.098	0.469	0.162
	Gemma 4	0.508	2.608	0.398	0.262	0.218	0.238	0.160	0.304	0.547	1.588	0.512	0.430	2.822	0.311	0.085	0.560	0.147
	Gemini 3.5	0.493	2.319	0.397	0.276	0.175	0.215	0.198	0.340	0.612	1.133	0.571	0.429	2.686	0.325	0.087	0.512	0.149
	GPT 5.6	0.500	2.863	0.383	0.273	0.182	0.218	0.191	0.340	0.640	1.284	0.527	0.428	3.429	0.327	0.096	0.510	0.162

Table 10. Comparison of frame-wise accuracy (fAcc) for each event class across top-line systems. Cascaded systems use Audio Flamingo 3 as the audio captioning model. The Ratio column indicates the proportion of frames belonging to each class within the dataset. Classes are sorted by ratio.

Ego4D Class	Ratio	Cascaded				End-to-end			
		Gemini 3.5	GPT 5.6	Gemma 4	Gemma 4 + SFT	Gemini 3.5	Qwen 3-o	Qwen 3-o + SFT	Qwen 2.5-o
productive	0.401	0.369	0.336	0.324	0.603	0.503	0.269	0.763	0.945
leisure	0.285	0.601	0.616	0.688	0.695	0.513	0.841	0.680	0.017
food	0.117	0.460	0.408	0.451	0.532	0.447	0.229	0.646	0.000
traveling	0.082	0.157	0.164	0.138	0.054	0.114	0.042	0.211	0.132
other	0.059	0.086	0.133	0.055	0.000	0.191	0.264	0.000	0.000
purchasing	0.056	0.098	0.072	0.088	0.232	0.434	0.348	0.720	0.067

EgoLife Class	Ratio	Cascaded				End-to-end			
		Gemini 3.5	GPT 5.6	Gemma 4	Gemma 4 + SFT	Gemini 3.5	Qwen 3-o	Qwen 3-o + SFT	Qwen 2.5-o
productive	0.643	0.173	0.145	0.129	0.127	0.262	0.013	0.136	0.970
leisure	0.268	0.712	0.778	0.777	0.849	0.418	0.857	0.671	0.013
purchasing	0.039	0.025	0.015	0.020	0.200	0.652	0.544	0.805	0.018
traveling	0.029	0.272	0.272	0.226	0.151	0.491	0.221	0.705	0.304
food	0.019	0.767	0.719	0.691	0.764	0.602	0.386	0.737	0.059
other	0.003	0.117	0.186	0.052	0.000	0.188	0.208	0.000	0.000

SINS Class	Ratio	Cascaded				End-to-end			
		Gemini 3.5	GPT 5.6	Gemma 4	Gemma 4 + SFT	Gemini 3.5	Qwen 3-o	Qwen 3-o + SFT	Qwen 2.5-o
sleeping	0.273	0.037	0.023	0.102	0.057	0.349	0.088	0.181	0.016
watching tv	0.226	0.663	0.511	0.391	0.762	0.774	0.722	0.948	0.760
absence	0.171	0.975	0.978	0.912	0.000	0.483	0.914	0.000	0.402
working	0.171	0.902	0.886	0.894	0.986	0.350	0.189	0.980	0.034
cooking	0.035	0.571	0.410	0.468	0.727	0.615	0.602	0.587	0.139
other	0.025	0.502	0.528	0.504	0.352	0.399	0.434	0.070	0.553
calling	0.020	0.602	0.794	0.671	0.124	0.613	0.270	0.190	0.033
eating	0.017	0.601	0.599	0.501	0.240	0.769	0.170	0.384	0.058
visit	0.013	0.786	0.723	0.806	0.276	0.250	0.179	0.735	0.000
toilet	0.011	0.197	0.168	0.177	0.000	0.040	0.000	0.000	0.010
dishwashing	0.011	0.263	0.286	0.312	0.379	0.512	0.138	0.000	0.000
vacuumcleaner	0.007	0.909	0.867	0.938	0.942	0.747	0.081	0.000	0.000
showering	0.007	0.778	0.729	0.843	0.785	0.713	0.348	0.879	0.318
dresssing	0.005	0.138	0.078	0.063	0.139	0.416	0.136	0.094	0.003
toothbrushing	0.003	0.893	0.825	0.895	0.248	0.621	0.000	0.000	0.441
shaving	0.003	0.934	0.944	0.933	0.850	0.694	0.582	0.870	0.000
getting dry	0.002	0.710	0.577	0.642	0.091	0.070	0.092	0.000	0.000